화학특강 (2009년 2학기)
제2장

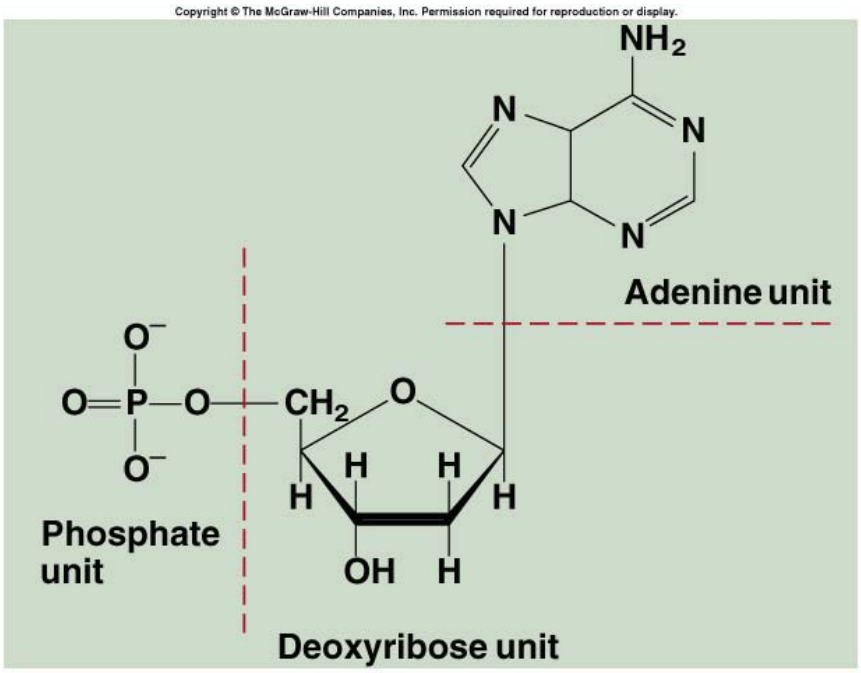
황현석

2009년 9월 14일

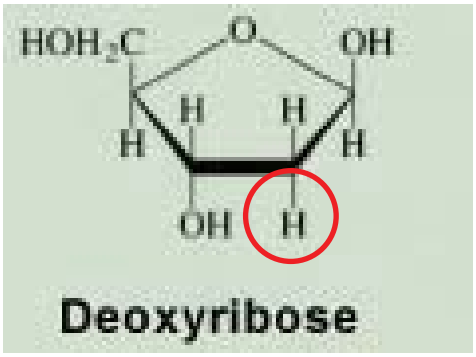


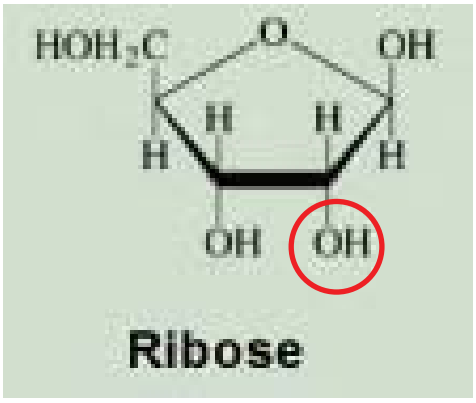
Structures of Deoxyribonucleic Acid (DNA) and Ribonucleic Acid (RNA)

- Structure of Nucleotide



- Deoxyribose and ribose



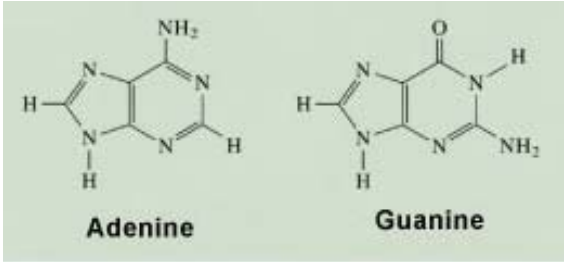


Chang, 25.4

KNU

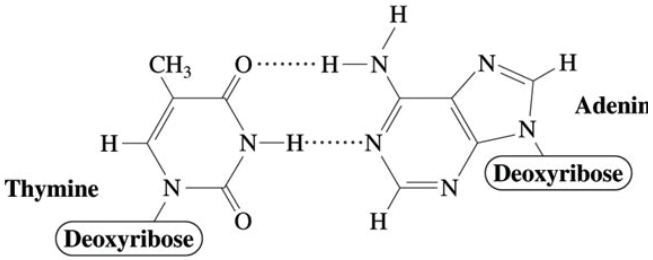
Structures of Deoxyribonucleic Acid (DNA) and Ribonucleic Acid (RNA)

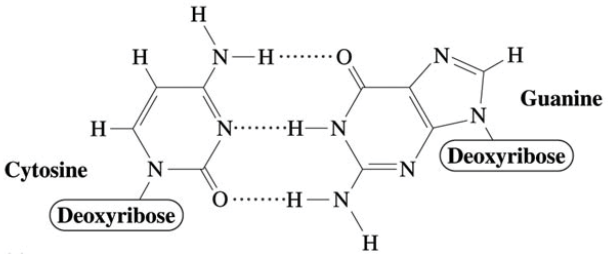
- Purine bases



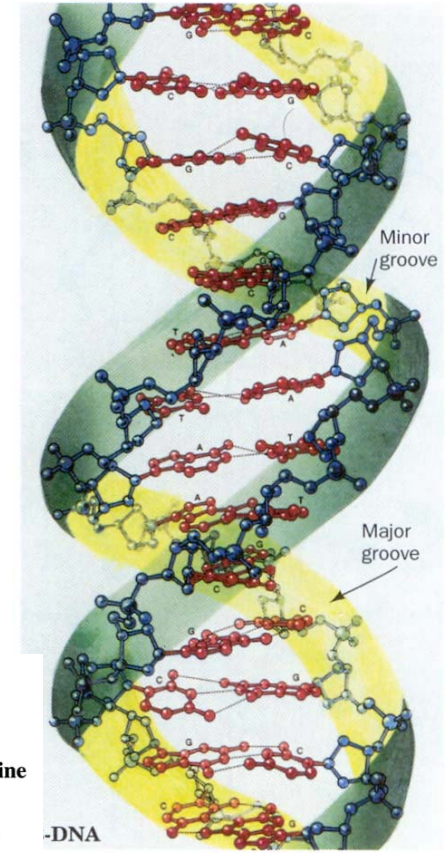
- Pyrimidine bases







- DNA double helix

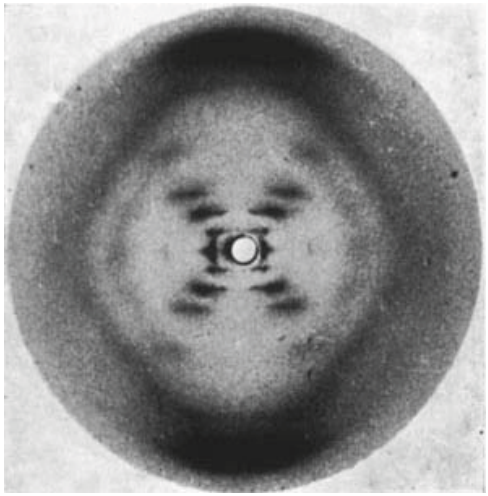


Chang, 25.4

History of DNA

- Chargaff's rules in the 1940's
  1. The amount of adenine (a purine) is equal to that of thymine (a pyrimidine); that is A = T, or A/T = 1.
  2. The amount of guanine (a purine) is equal to that of cytosine (a pyrimidine); that is G = C, or G/C = 1.
  3. The total number of purine bases is equal to the total number of pyrimidine bases; that is, A + G = T + C .

- X-ray diffraction photograph by R. Franklin





Chang, 25.4

KNU

History of DNA

- Nature paper by Watson & Crick

No. 4356 April 25, 1953 NATURE
737

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

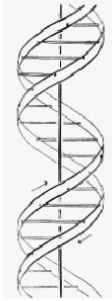
We wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

A structure for nucleic acid has already been proposed by Pauling and Corey[1]. They kindly made their manuscript available to us in advance of publication. Their model consists of three intertwined chains, with the phosphates near the fibre axis, and the bases on the outside. In our opinion, this structure is unsatisfactory for two reasons:

(1) We believe that the material which gives the X-ray diagrams is the salt, not the free acid. Without the acidic hydrogen atoms it is not clear what forces would hold the structure together, especially as the negatively charged phosphates near the axis will repel each other. (2) Some of the van der Waals distances appear to be too small.

Another three-chain structure has also been suggested by Fraser (in the press). In his model the phosphates are on the outside and the bases on the inside, linked together by hydrogen bonds. This structure as described is rather ill-defined, and for this reason we shall not comment on it.

We wish to put forward a radically different structure for the salt of deoxyribose nucleic acid. This structure has two helical chains each coiled round the same axis (see diagram). We have made the usual chemical assumptions, namely, that each chain consists of phosphate diester groups joining β-D-deoxyribofuranose residues with 3',5' linkages. The two chains (but not their bases) are related by a dyad perpendicular to the fibre axis. Both chains follow righthanded helices, but owing to the dyad the sequences of the atoms in the two chains run in opposite directions.



This figure is purely diagrammatic. The two ribbons symbolize the two phosphate—sugar chains, and the horizontal rods the pairs of bases holding the chains together. The vertical line marks the fibre axis

Each chain loosely resembles Furberg's[2] model No. 1; that is, the bases are on the inside of the helix and the phosphates on the outside. The configuration of the sugar and the atoms near it is close to Furberg's 'standard configuration', the sugar being roughly perpendicular to the attached base. There is a residue on each chain every 3·4 A. in the z-direction. We have assumed an angle of 36° between adjacent residues in the same chain, so that the structure repeats after 10 residues on each chain, that is, after 34 A. The distance of a phosphorus atom from the fibre axis is 10 A. As the phosphates are on the outside, cations have easy access to them.

The structure is an open one, and its water content is rather high. At lower water contents we would expect the bases to tilt so that the structure could become more compact.

The novel feature of the structure is the manner in which the two chains are held together by the purine and pyrimidine bases. The planes of the bases are perpendicular to the fibre axis. They are joined together in pairs, a single base from one chain being hydrogen-bonded to a single base from the other chain, so that the two lie side by side with identical z-co-ordinates. One of the pair must be a purine and the other a pyrimidine for bonding to occur. The hydrogen bonds are made as follows: purine position 1 to pyrimidine position 1; purine position 6 to pyrimidine position 6.

If it is assumed that the bases only occur in the structure in the most plausible tautomeric forms (that is, with the keto rather than the enol configurations) it is found that only specific pairs of bases can bond together. These pairs are: adenine (purine) with thymine (pyrimidine), and guanine (purine) with cytosine (pyrimidine).

In other words, if an adenine forms one member of a pair, on either chain, then on these assumptions the other member must be thymine; similarly for guanine and cytosine. The sequence of bases on a single chain, does not appear to be restricted in any way. However, if only specific pairs of bases can be formed, it follows that if the sequence of bases on one chain, is given, then the sequence on the other chain is automatically determined.

It has been found experimentally[3,4] that the ratio of the amounts of adenine to thymine, and the ratio of guanine to cytosine, are always very close to unity for deoxyribose nucleic acid.

It is probably impossible to build this structure with a ribose sugar in place of the deoxyribose, as the extra oxygen atom would make too close a van der Waals contact.

The previously published X-ray data[5,6] on deoxyribose nucleic acid are insufficient for a rigorous test of our structure. So far as we can tell, it is roughly compatible with the experimental data, but it must be regarded as unproved until it has been checked against more exact results. Some of these are given in time following, communications. We were not aware of the details of the results presented there when we devised our structure, which rests mainly though not entirely on published experimental data and stereo-chemical arguments.

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

Full details of the structure, including the conditions assumed in building it, together with a set of co-ordinates for the atoms, will be published elsewhere.

We are much indebted to Dr. Jerry Donohue for constant advice and criticism, especially on interatomic distances. We have also been stimulated by a knowledge of the general nature of the unpublished experimental results and ideas of Dr. *M.* H. F. Wilkins, Dr. R. E. Franklin and their co-workers at King's College, London. One of us (J.D.W.) has been aided by a fellowship from the National Foundation for Infantile Paralysis.

J.D. WATSON
F.H. C. CRICK

Medical Research Council Unit for the Study of the Molecular Structure of Biological Systems, Cavendish Laboratory, Cambridge. April 2.

[1] Pauling, L., and Corey, R. B. nature, 171, 346 (1953); Proc. U.S. Nat Acad. Sci., 39, 84 (1953).
[2] Furberg, S., Acta Chem. Scand., 6, 634 (1952).
[3] Chargaff, E., for references see Zamenhof, S., Brawerman, G., and Chargaff, E., Biochim. et Biophys. Acta, 9 402 (1952).
[4] Wyatt, G.R. J. Gen. Physiol., 36 201 (1952).
[5] Astbury, W.T., Symp. Soc. Exp. Biol. 1, Nucleic Acid, 66 (Camb. Univ. Press, 1947)
[6] Wilkins, M. H. F. and Randall, J. T. Biochim. et. Biophys. Acta, 10, 102 (1953).

Watson & Crick, Nature 171, 737 (1953)



The Nobel Prize in Physiology or Medicine 1962

"for their discoveries concerning the molecular structure of nucleic acids and its significance for information transfer in living material"



Francis Harry Compton Crick

1/3 of the prize

United Kingdom

MRC Laboratory of Molecular Biology
Cambridge, United Kingdom



James Dewey Watson

1/3 of the prize

USA

Harvard University
Cambridge, MA, USA



Maurice Hugh Frederick Wilkins

1/3 of the prize

United Kingdom and New Zealand

London University
London, United Kingdom

Amino Acids

- **Amino acids**: the basic structural units of proteins. An amino acid is a compound that contains at least one amino group (-NH2) and at least one carboxyl group (-COOH)

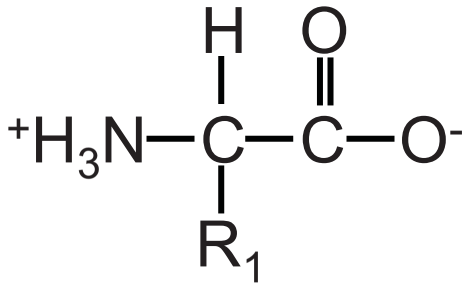
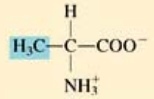
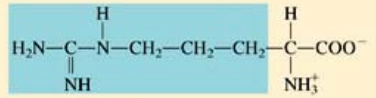
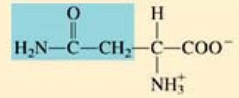
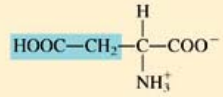
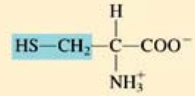
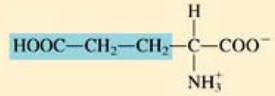
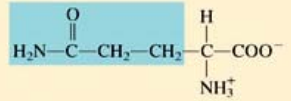
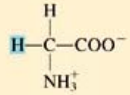
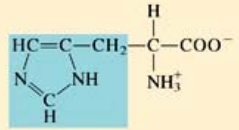
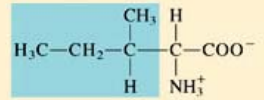


TABLE 25.1 The 20 Amino Acids Essential to Living Organisms*

Name	Abbreviation	Structure
Alanine	Ala	
Arginine	Arg	
Asparagine	Asn	
Aspartic acid	Asp	
Cysteine	Cys	
Glutamic acid	Glu	
Glutamine	Gln	
Glycine	Gly	
Histidine	His	
Isoleucine	Ile	

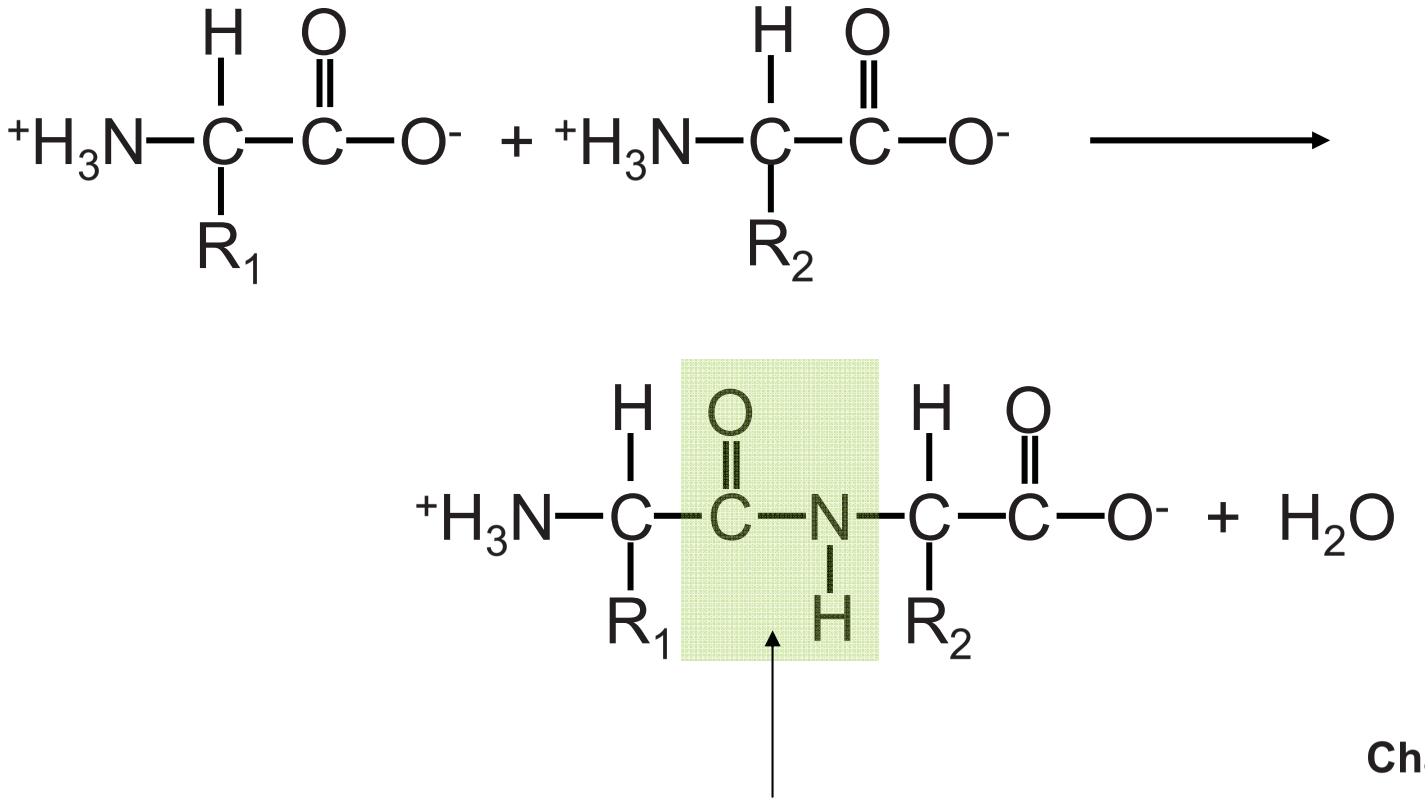
(Continued)

*The shaded portion is the R group of the amino acid.

Chang, 25.4

Proteins

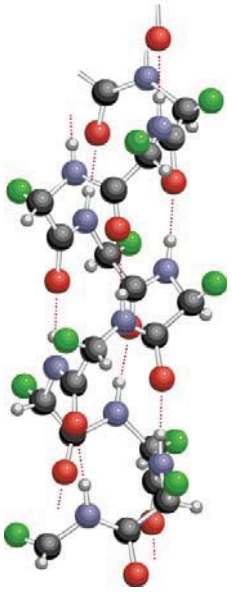
- **Proteins**: A polymer of amino acids through peptide bonds.



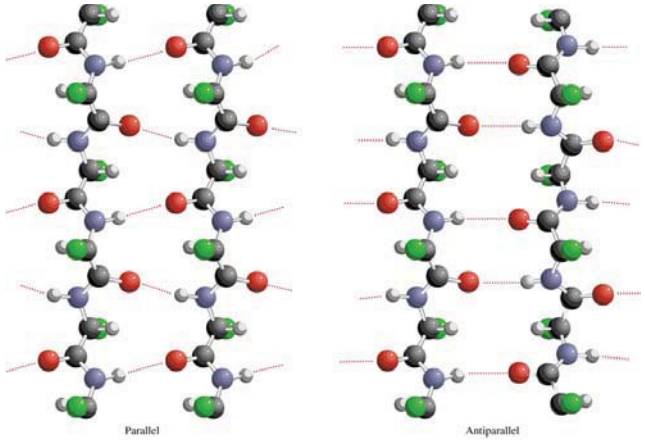
Chang, 25.4

Protein Structure

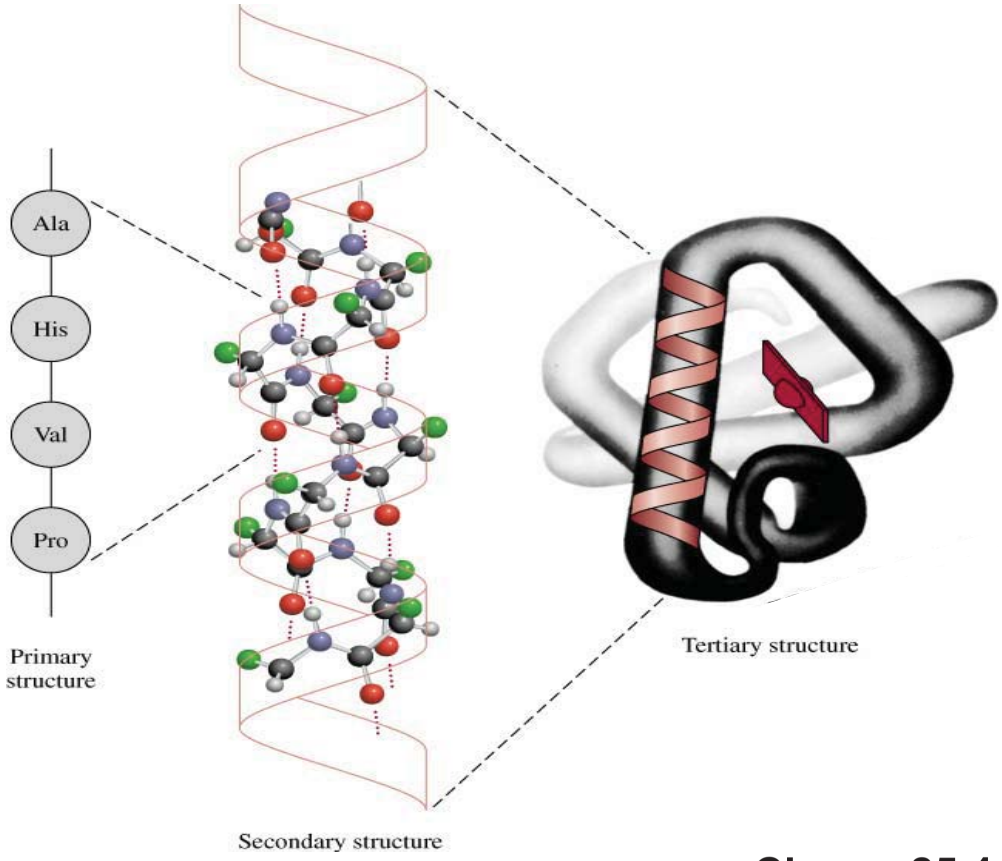
- α helix



- β sheet



- Primary, secondary, and tertiary structures

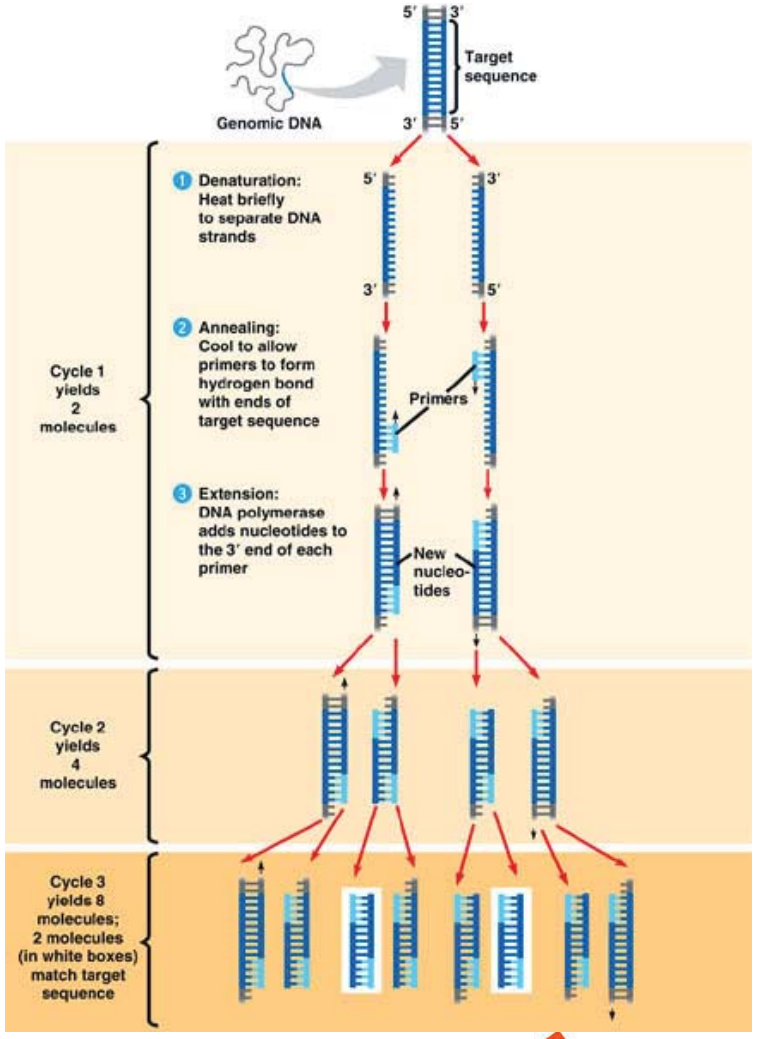


Chang, 25.4

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Polymerase Chain Reaction (PCR)

- **PCR**
  - A technique to amplify a piece of DNA
  - DNA template (target), two primers, DNA polymerase (Taq polymerase), deoxinucleoside triphosphates.
- Procedure
  - Initialization: heating to 94-96°C
  - Denaturation: yielding ssDNAs
  - Annealing: lowering to 50-65°C and annealing primers to ssDNAs
  - Elongation: inducing DNA polymerases to synthesize a new DNA strand
- Applications of PCR
  - Isolation of genomic DNA
  - DNA sequencing
  - Forensic analysis
  - Disease diagnosis



Polymerase Chain Reaction (PCR)

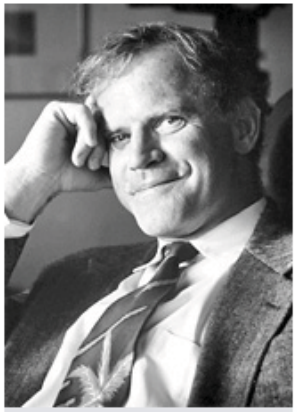


The Nobel Prize in Chemistry 1993

"for contributions to the developments of methods within DNA-based chemistry"

"for his invention of the polymerase chain reaction (PCR) method"

"for his fundamental contributions to the establishment of oligonucleotide-based, site-directed mutagenesis and its development for protein studies"

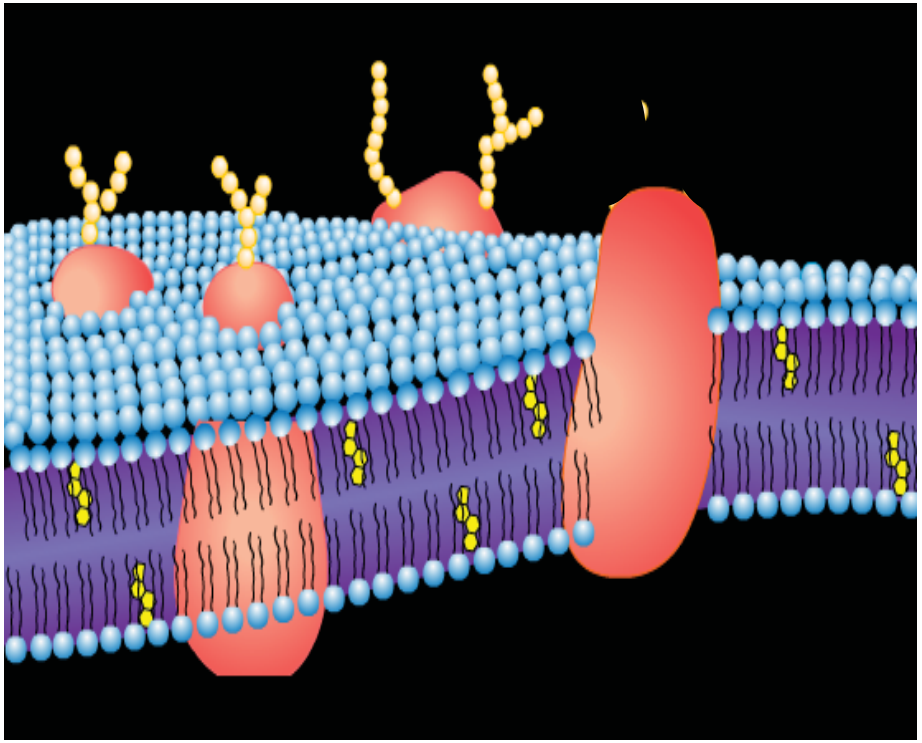


Kary B. Mullis

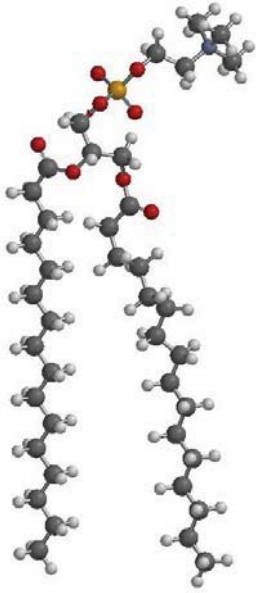


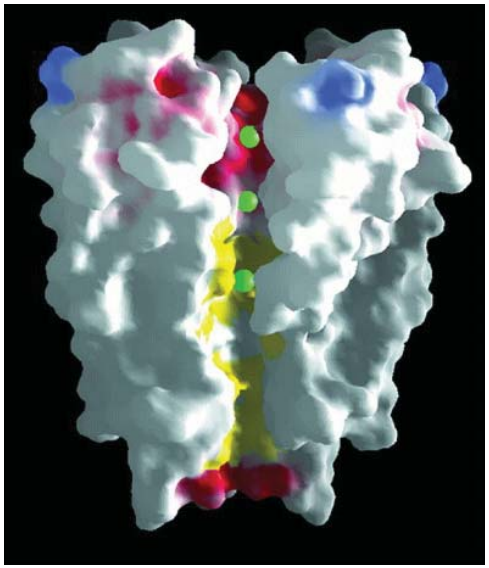
Michael Smith

What are ion channels?



Voet and Voet, Biochemistry (3rd. Ed.)





KcsA ion channel
Science **280,** 69 (1998)

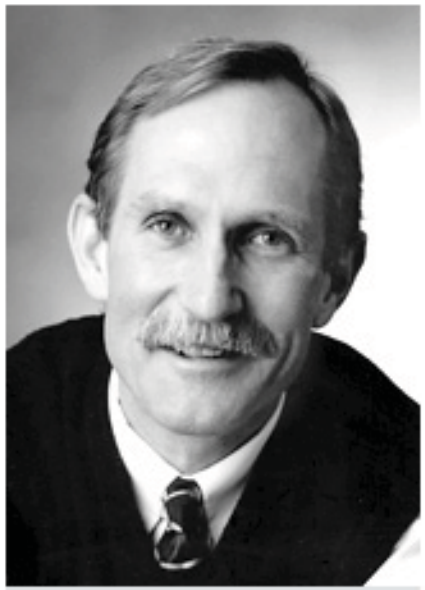


Chemistry

The Nobel Prize in Chemistry 2003

"for discoveries concerning channels in cell membranes"

"for the discovery of water channels"



Peter Agre

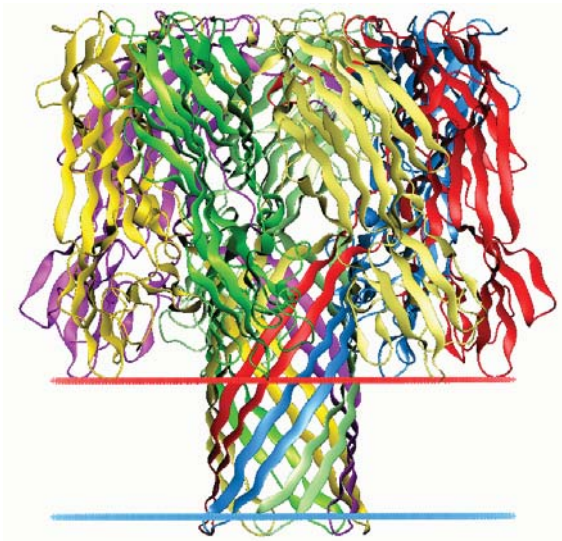
"for structural and mechanistic studies of ion channels"

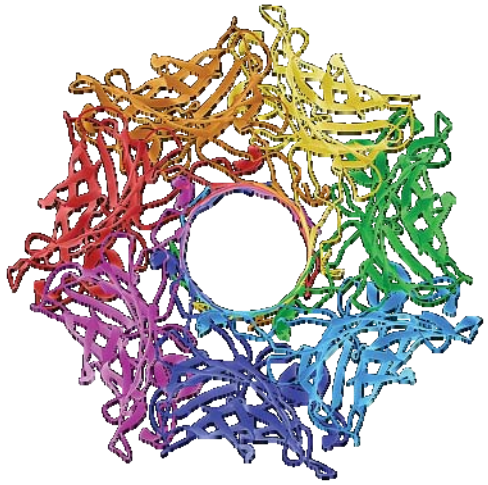


Roderick MacKinnon

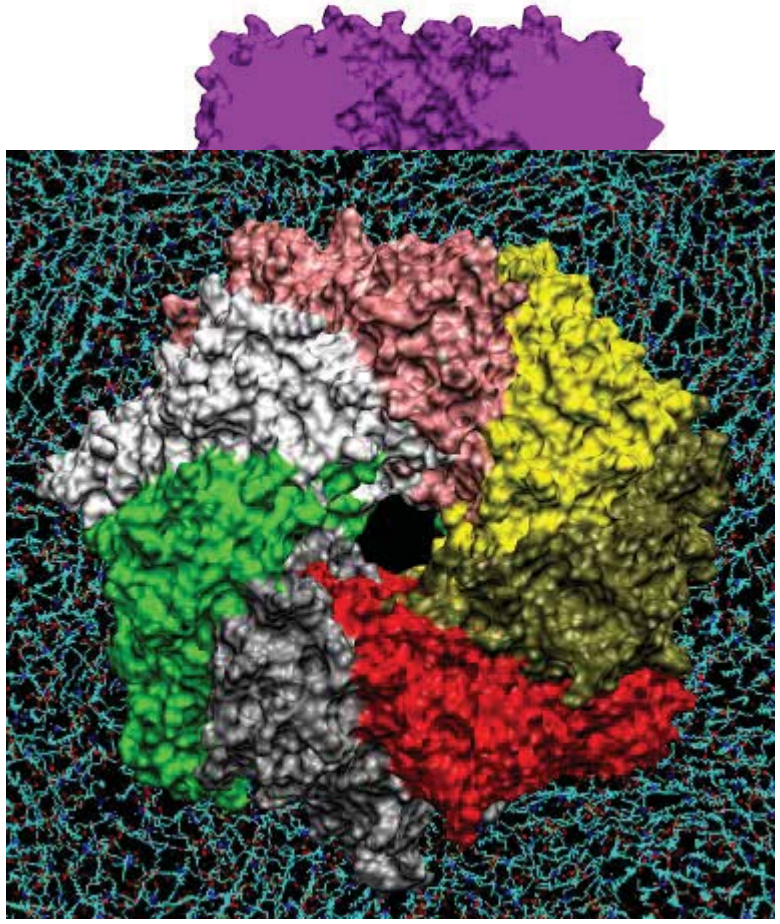
DNA translocation

- Alpha hemolysin: an ion channel

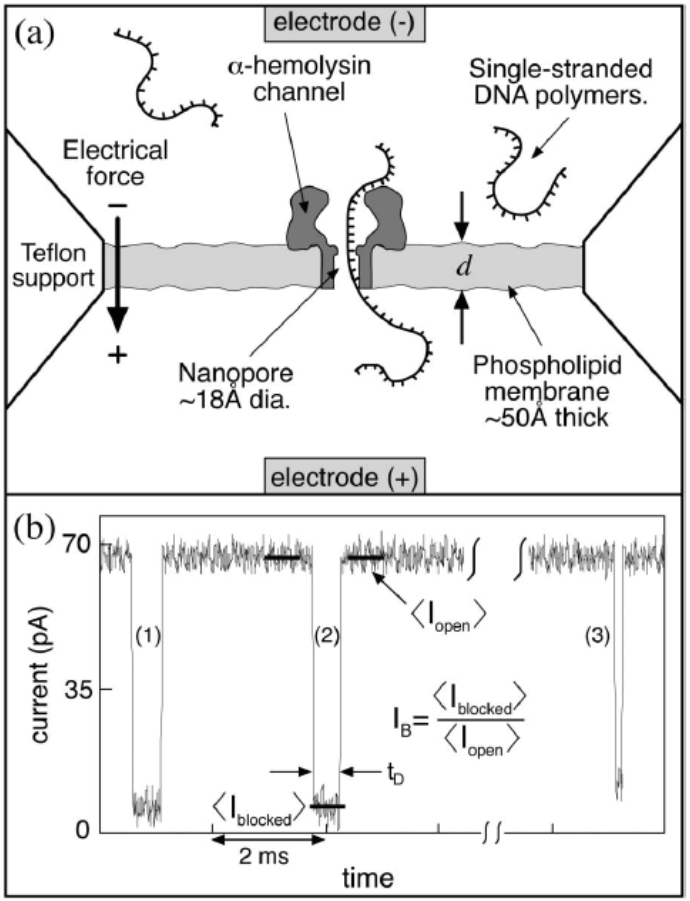




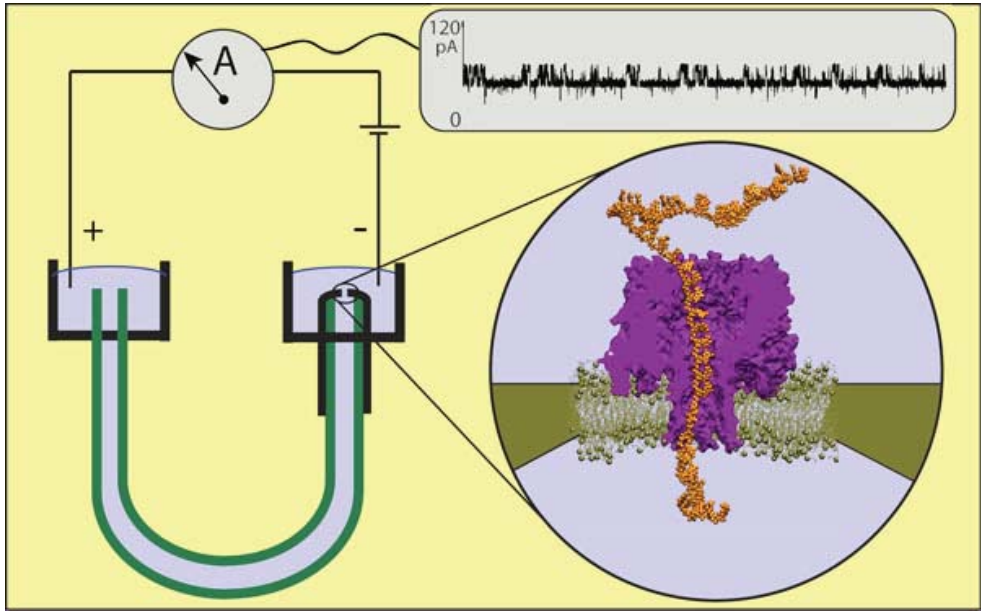
- Alpha hemolysin inside a lipid bilayer



DNA translocation

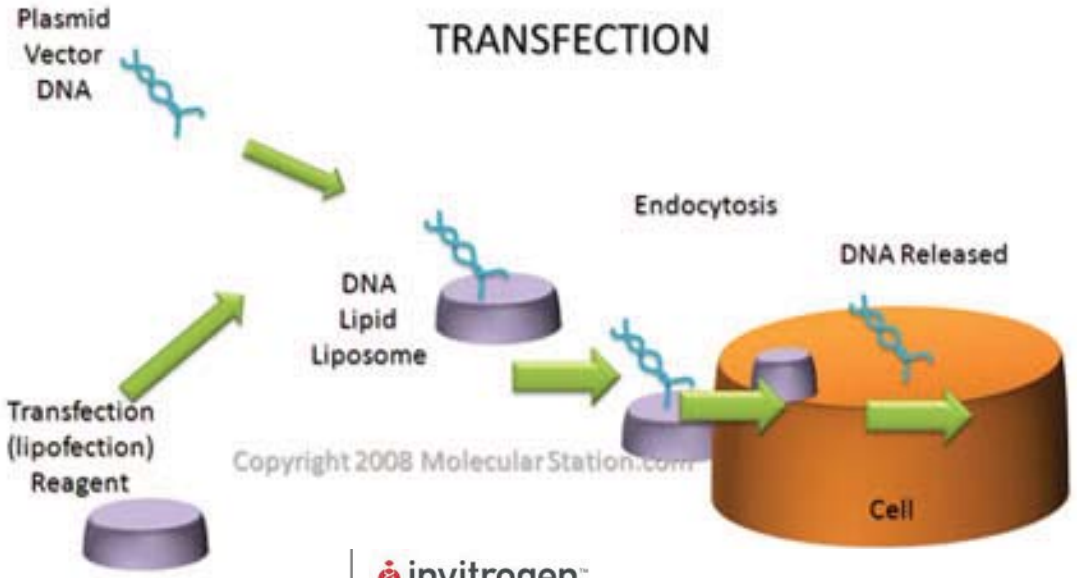


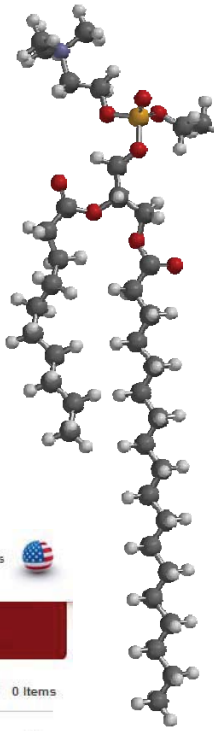
Miller et al., PRL **86**, 3435 (2001)

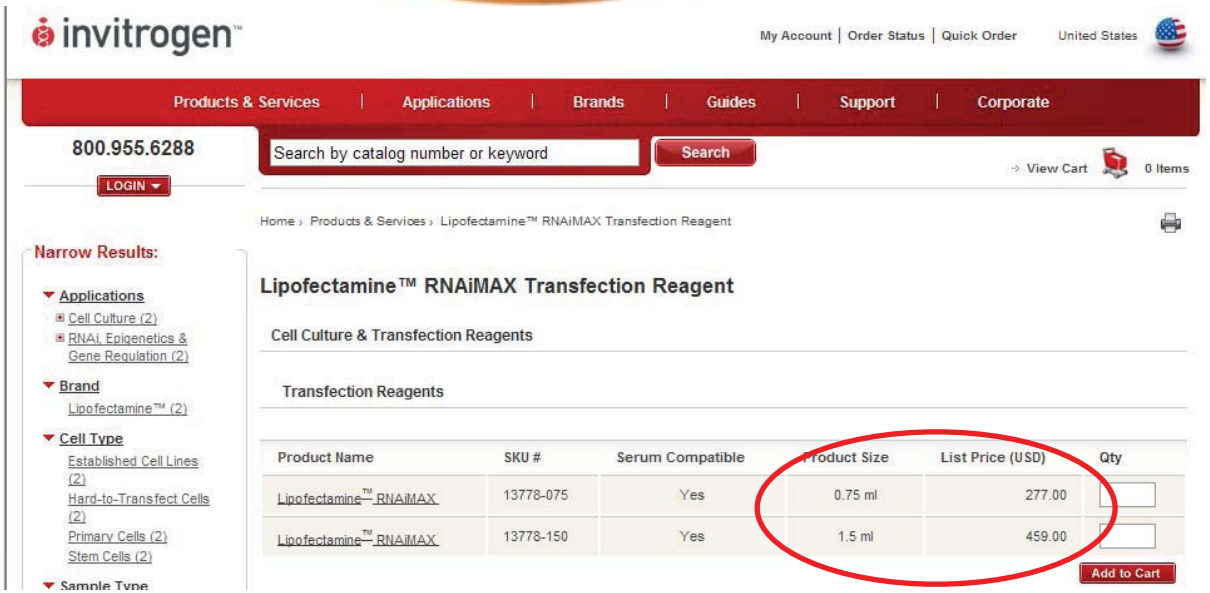


Schulten et al., PNAS **102**, 12377 (2005)

DNA Transfection

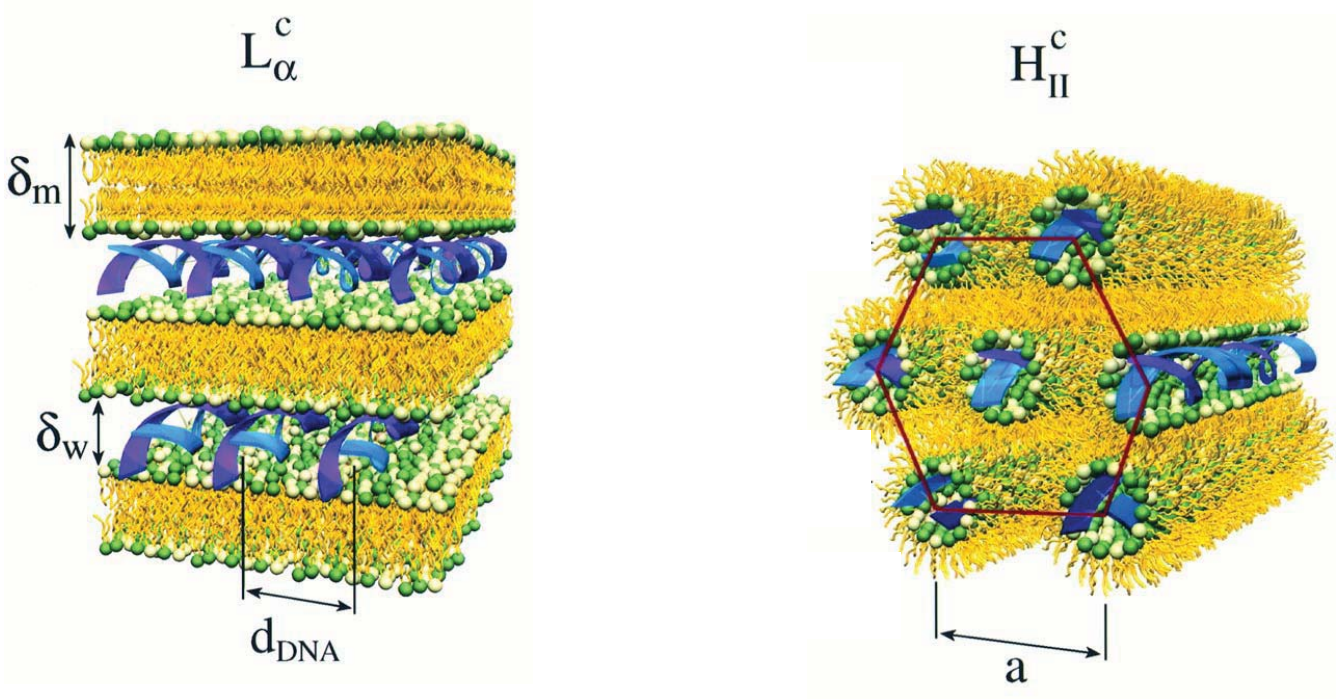






Lipid phases in lipid-DNA complex (=lipoplex)

- Lamellar(L_α^c) and hexagonal(H_{II}^c) phases.



(Science, **281**, 78 (1998))

Beer-Lambert Law

- Derivation of Beer-Lambert Law

$$dI = -\kappa[J]Idl \Rightarrow \frac{dI}{I} = -\kappa[J]dl$$

$$\int_{I_0}^{I} \frac{dI}{I} = -\kappa[J]\int_0^l dl \Rightarrow \ln\frac{I}{I_0} = -\kappa[J]l$$

$$\ln\frac{I}{I_0} = -\kappa[J]l \Rightarrow \log\frac{I}{I_0} = -\log e\kappa[J]l = -\varepsilon[J]l$$

- Absorbance

$$\boxed{A = \log\frac{I_0}{I} = \varepsilon[J]l}$$

Atkins, Physical Chemistry (8th ed.) 432-433

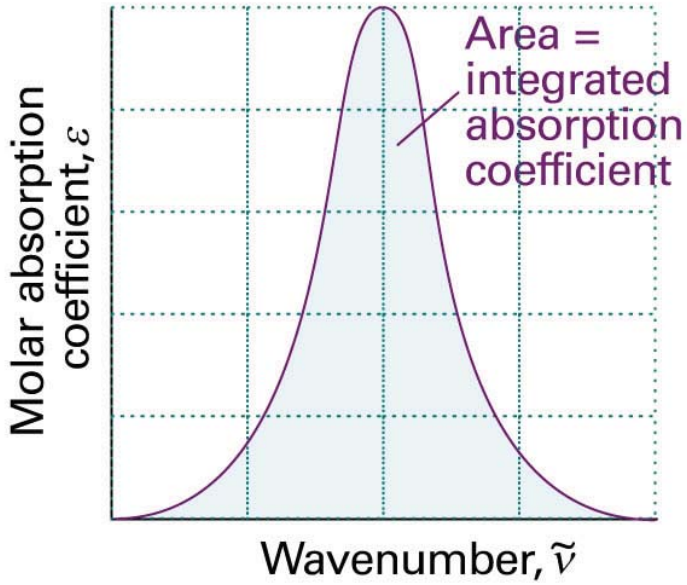
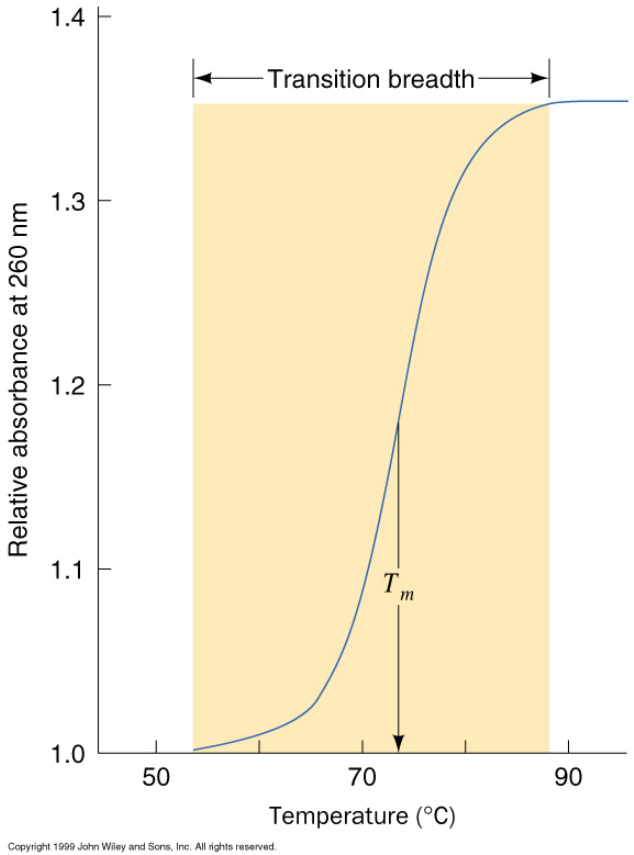


Fig. 13.4 The integrated absorption coefficient of a transition is the area under a plot of the molar absorption coefficient against the wavenumber of the incident radiation.

Hyperchromicity of DNA

- Hyperchromicity of DNA



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- Extinction coefficients of ss- and dsDNA at 260 nm
  1. For ssDNA: 0.027 $(\mu g/ml)^{-1}cm^{-1}$
  2. For dsDNA: 0.020 $(\mu g/ml)^{-1}cm^{-1}$

Colors and Complementary Colors

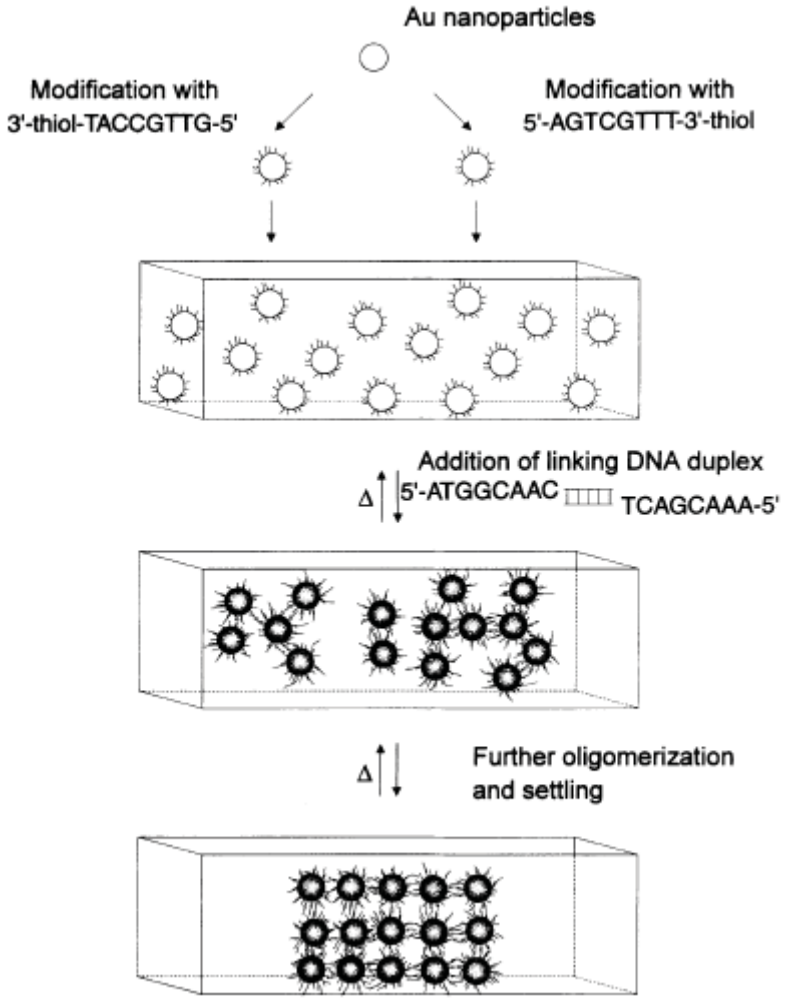
The colors of the visible light

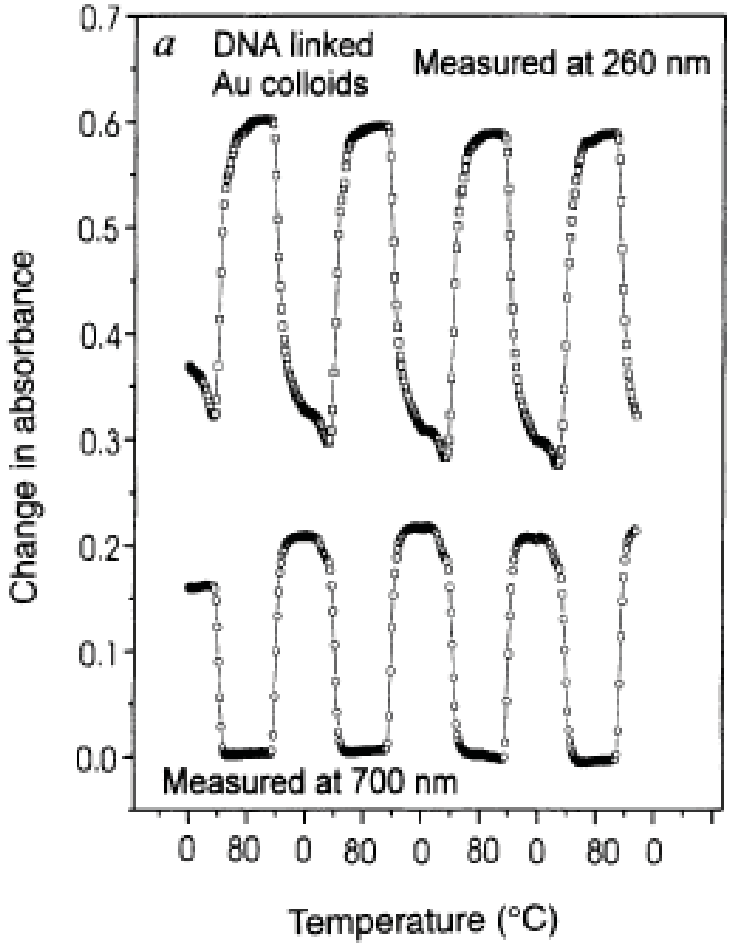
color	wavelength interval	frequency interval
red	700–635 nm	430–480 THz
orange	635–590 nm	480–510 THz
yellow	590–560 nm	510–540 THz
green	560–490 nm	540–610 THz
blue	490–450 nm	610–670 THz
violet	450–400 nm	670–750 THz

Complementary colors



Selective Colorimetric Detection of DNA

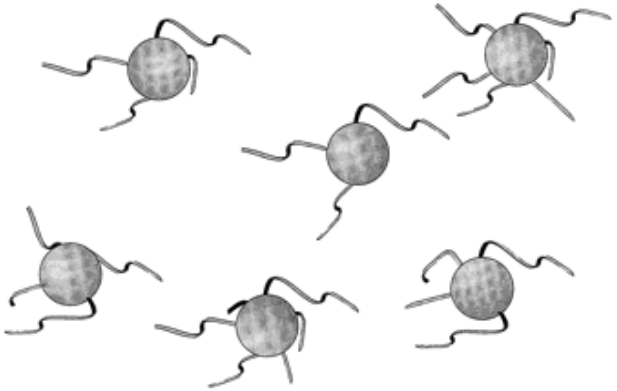




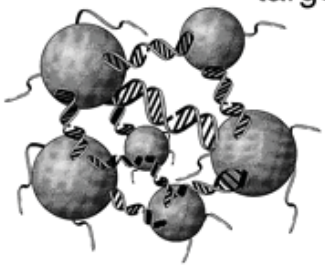
Mirkin et al., Nature 382, 607 (1996)

Selective Colorimetric Detection of DNA

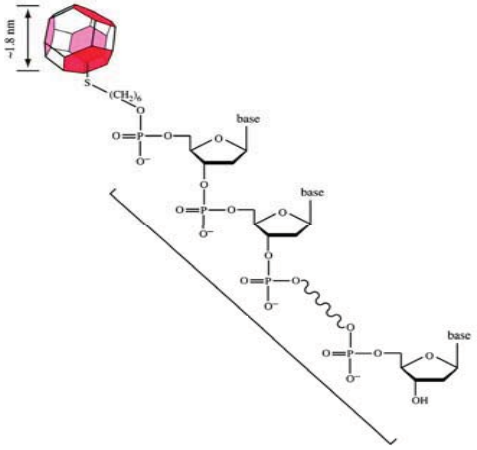
높은 온도 (58.5°C 이상)

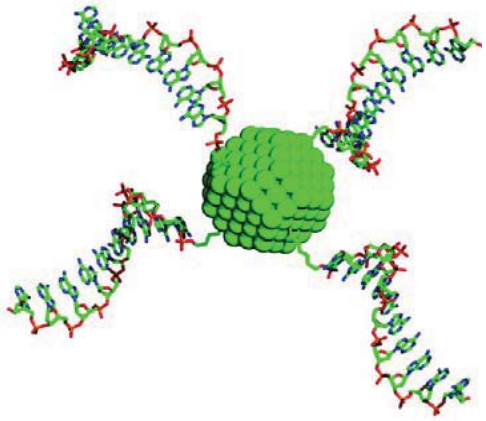


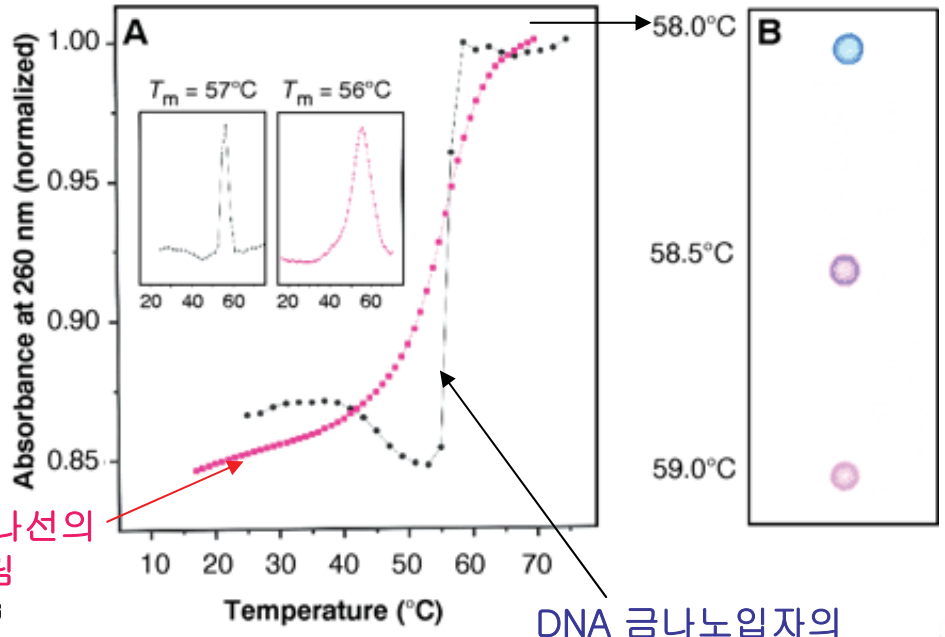
n Polynucleotide targets



낮은 온도 (58.5°C 이하)





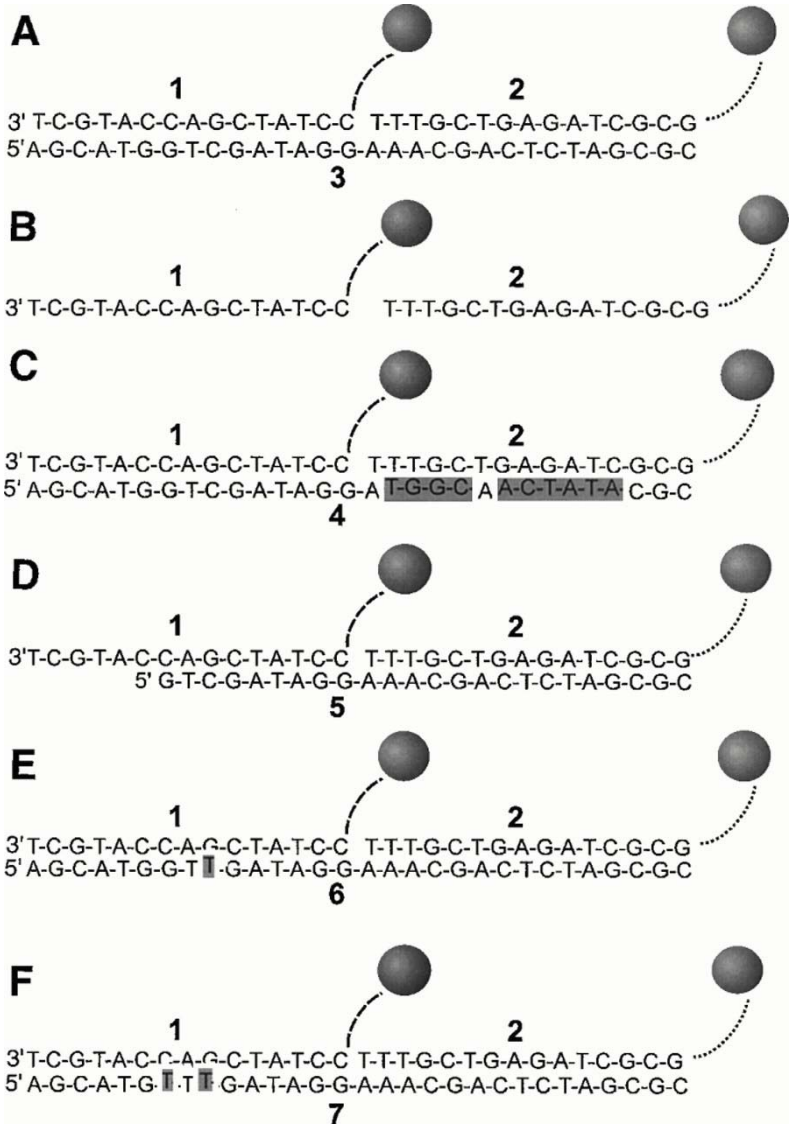


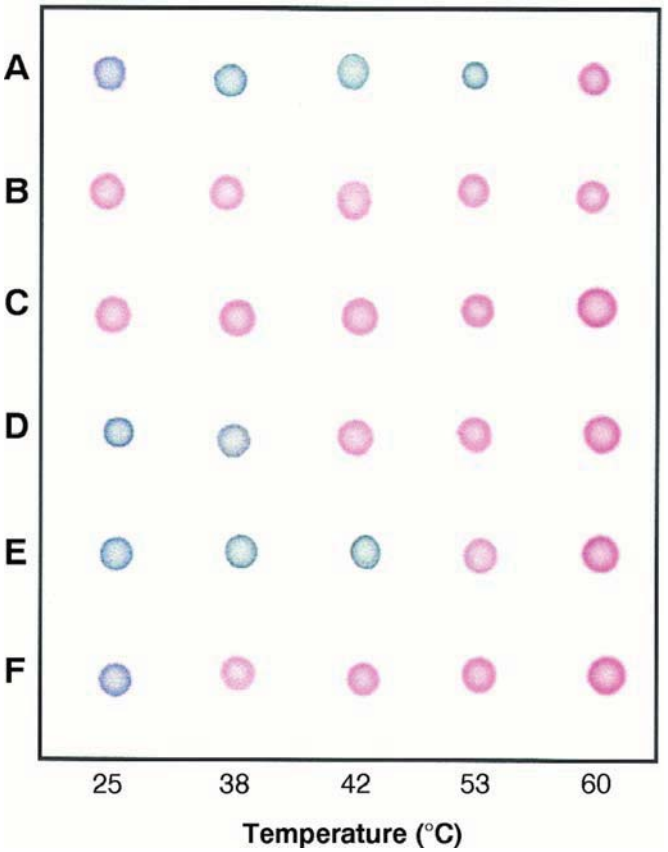
일반 DNA 이중나선의 상전이 그림

DNA 금나노입자의 상전이 그림

Mirkin et al., Science 277, 1078 (1997)

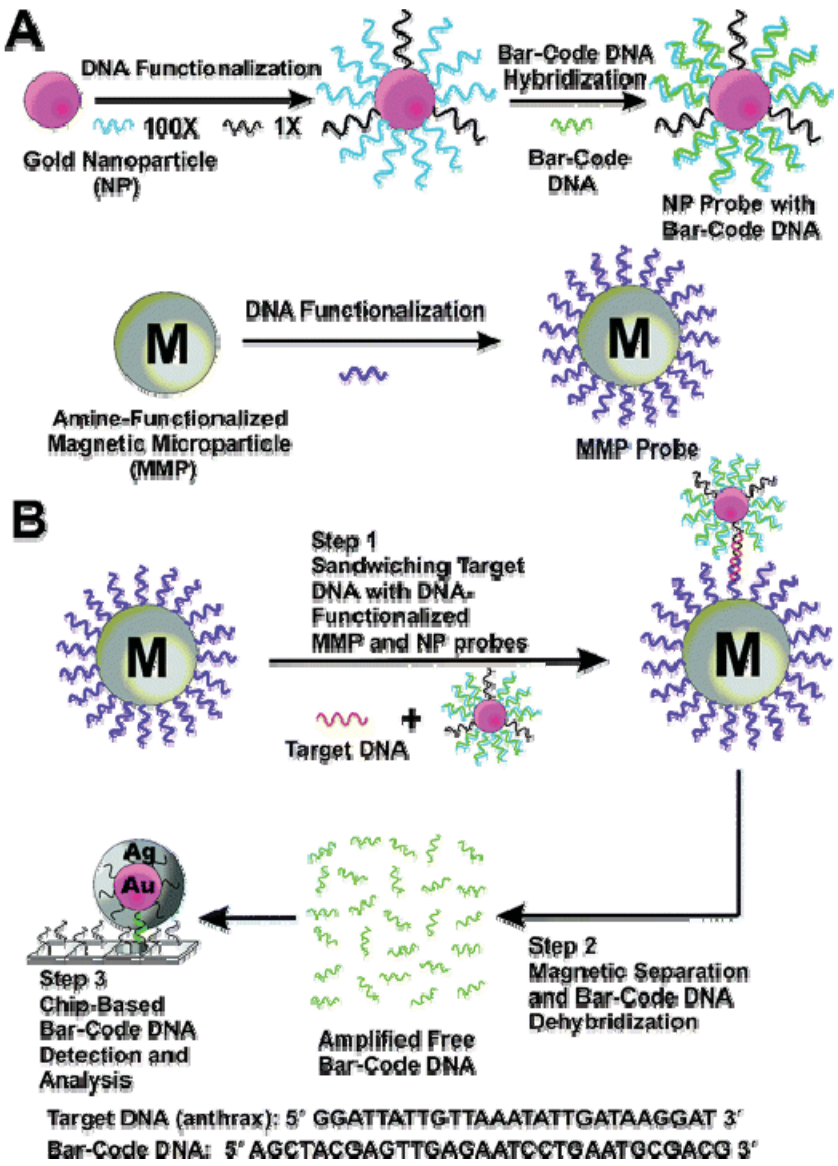
Selective Colorimetric Detection of DNA

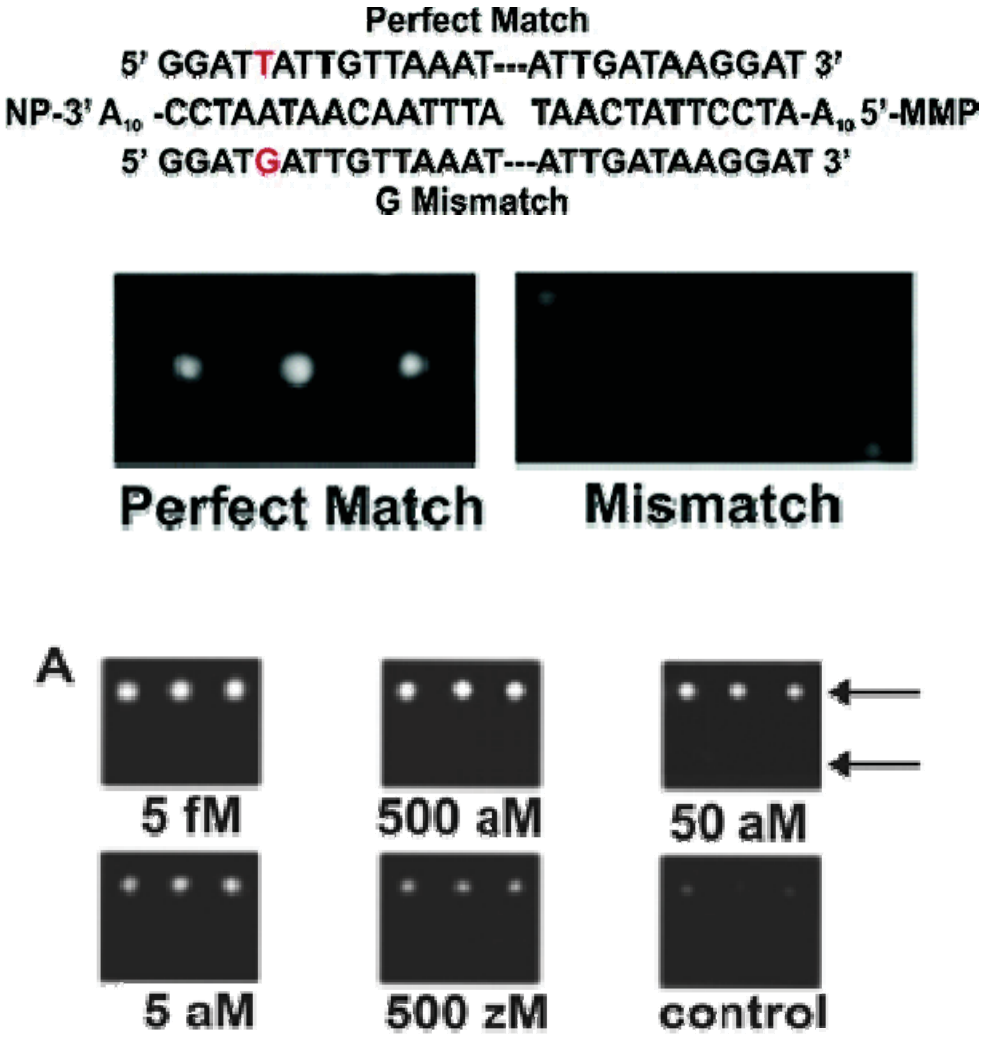




Mirkin et al., Science 277, 1078 (1997)

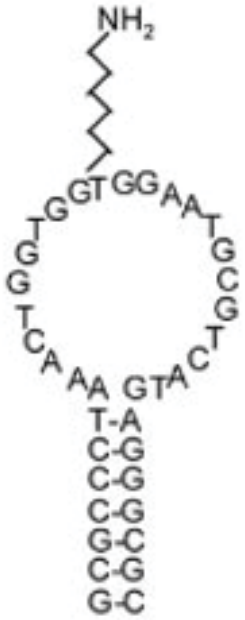
Bio-Bar-Code-Based DNA Detection

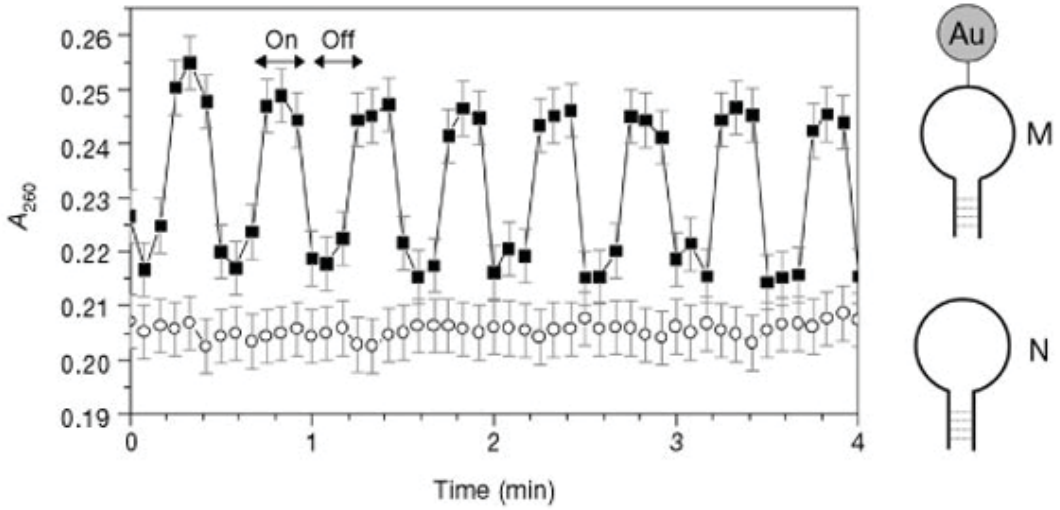




Mirkin et al., JACS **126**, 5932 (2004)

Remote Electronic Control of DNA Hybridization through Nanoscale Antenna

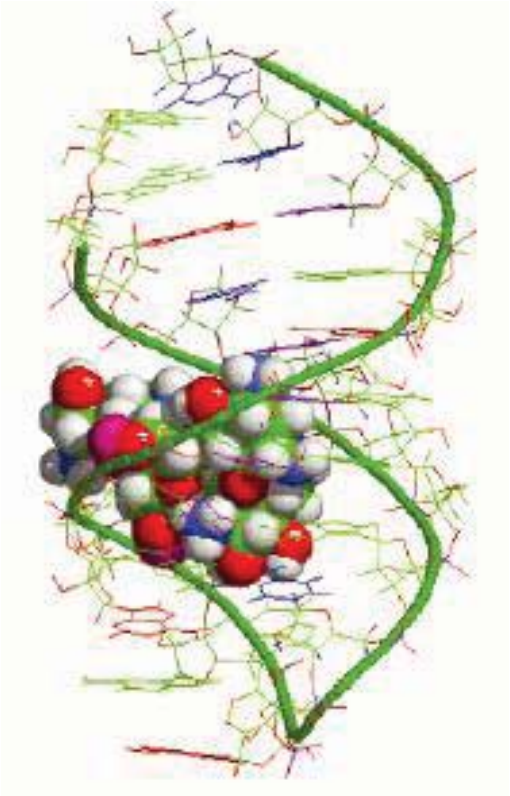


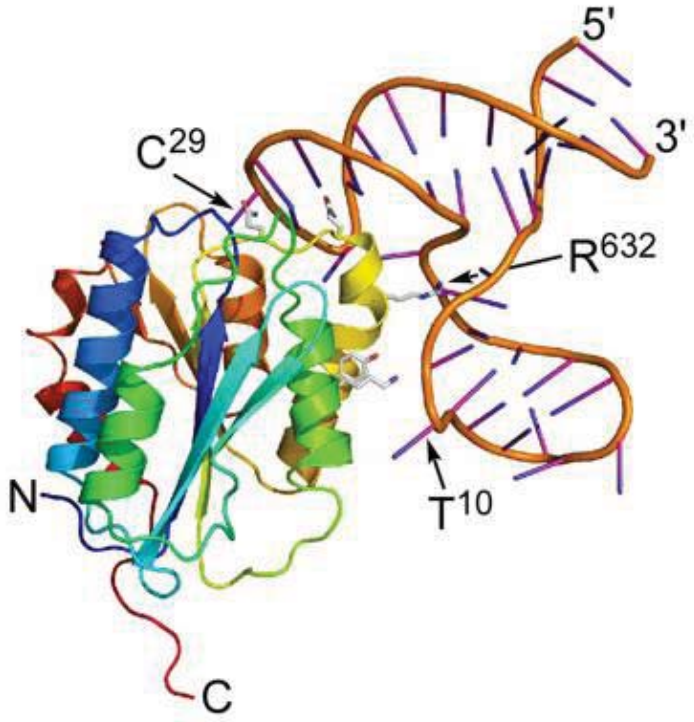


Hamad-Schifferli et al., Nature **415**, 152 (2002)

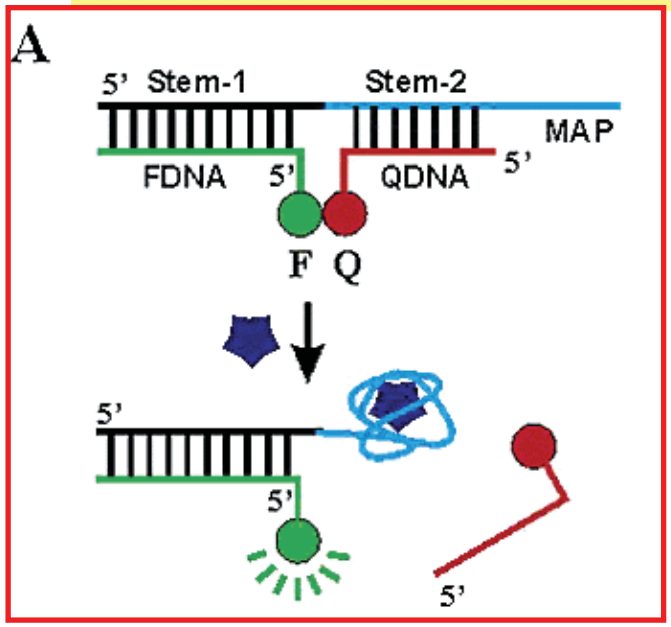
Aptamer

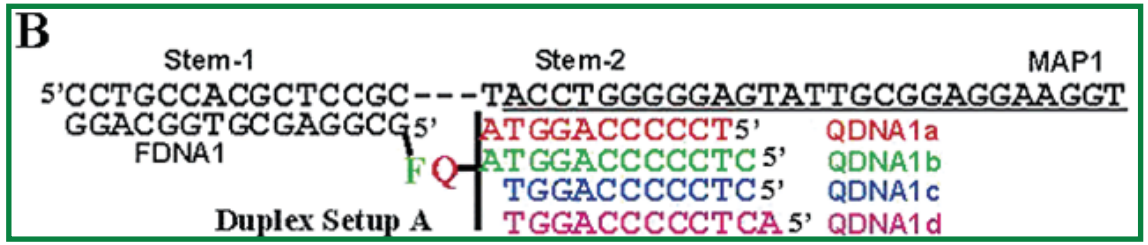
Aptamer: 핵산, 단백질, 금속이온과 같이 생체 내의 주요물질에 높은 결합력과 선택성을 가지고 결합하는 단일사슬 핵산 (single-stranded DNA)이나 단백질. 단백질로 이루어진 항체와는 달리 화학적, 물리적, 생물학적 안정성이 크다.
(aptamer means “to fit” in Greek)

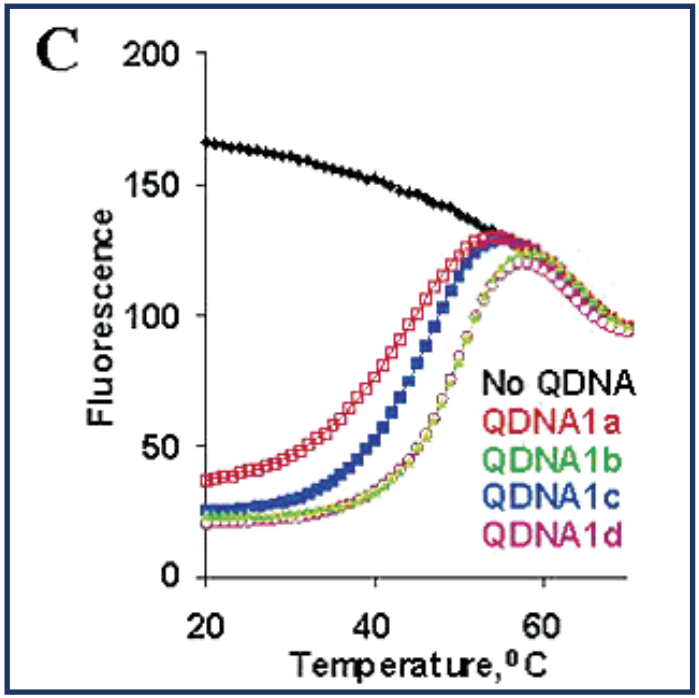


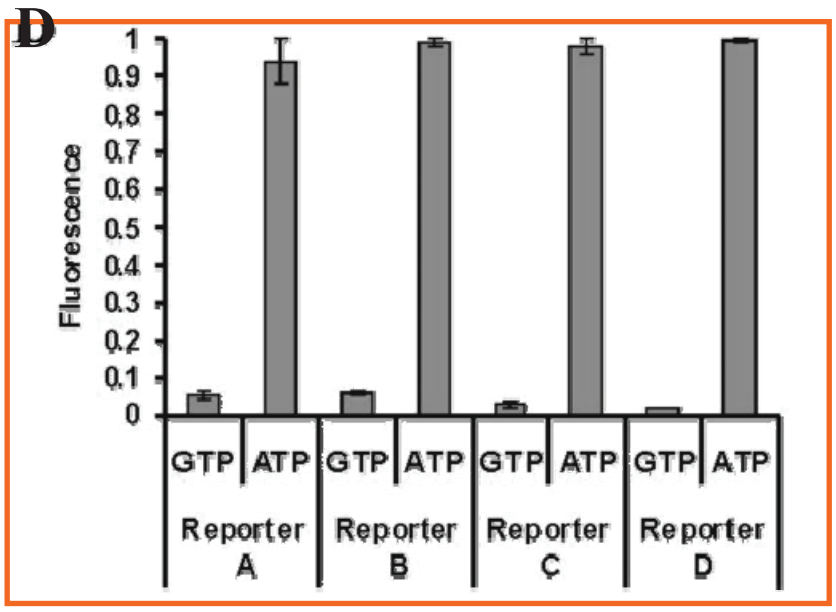


Colorimetric Sensing of Adenosine Based on Aptamer



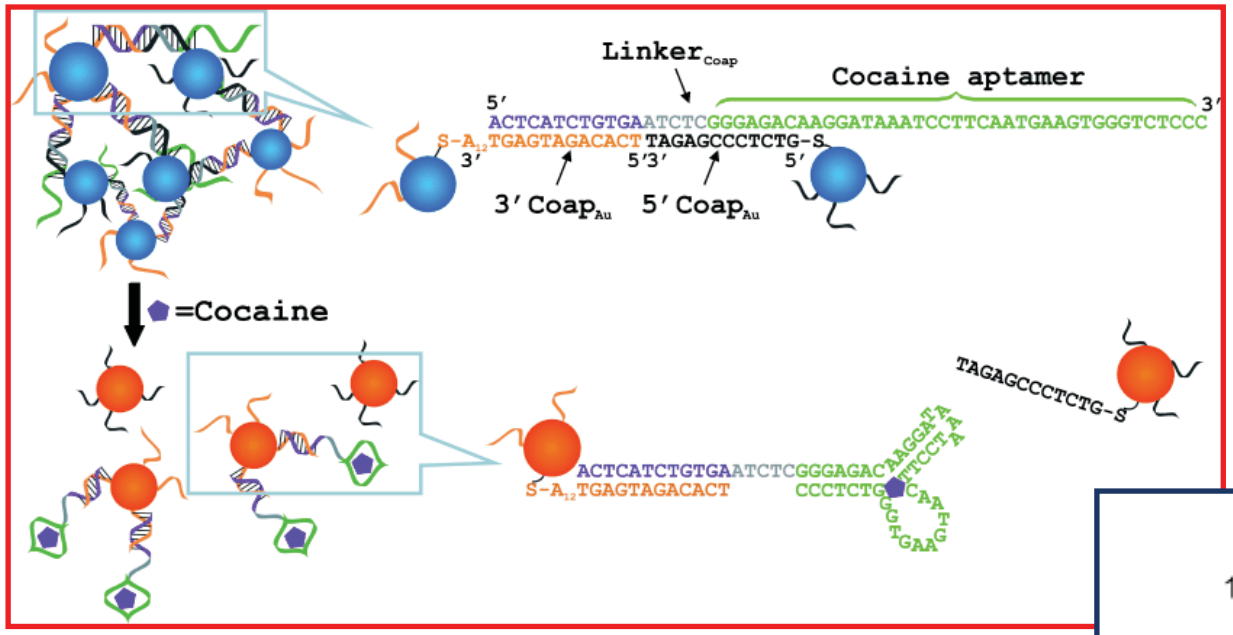


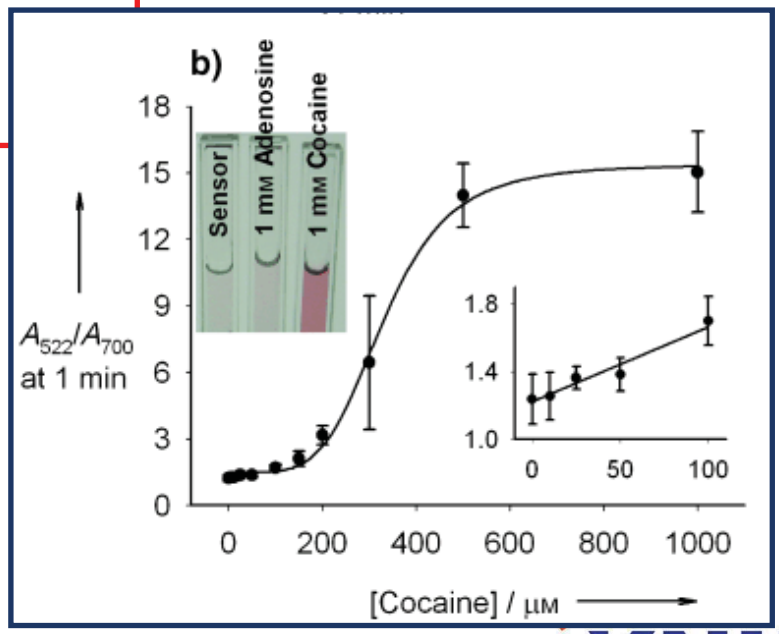




Nutiu and Li, JACS 125, 4771 (2003)

Colorimetric Sensing of Cocaine Based on Aptamer





DNA 금나노입자가 뭉쳐있을 때 → 700 nm 흡수, 파랑 또는 보라색

DNA 금나노입자가 떨어져 있을 때 → 522 nm 흡수, 붉은색

Liu and Lu, Angew. Chem. Int. Ed. **45**, 90 (2006)